

overview

What is data structure?

DATA + INSTRUCTIONS = PROGRAM

- Any computer software deals with **data**
 - **Organize Store Process Use it!**
 - **Facebook data:** Your account details (name, email, password), your friends list, your images and videos, your posts (replies, likes), Ads, groups, attached files ... etc
 - **Dictionary:** There are many thousands of words? How to store the words? Sorted? In an array? What about common **prefix** between words (**autopilot**, **autobiography**, **automobile**)? Search for a word that starts with “**univ**”?
 - **Google maps:** I want to go From School to Home? From Cairo to London? How to store these **locations/path** ? How to efficiently find the path between 2 points?

Why data structure?

Menu

Faculty of Computers and Information

Bank Masr, Egypt

Leave now

OPTIONS

Send directions to your phone

via 6th of October Bridge

Fastest route, despite the usual traffic

14 min

5.8 km

DETAILS

via El Tahrir

The usual traffic

18 min

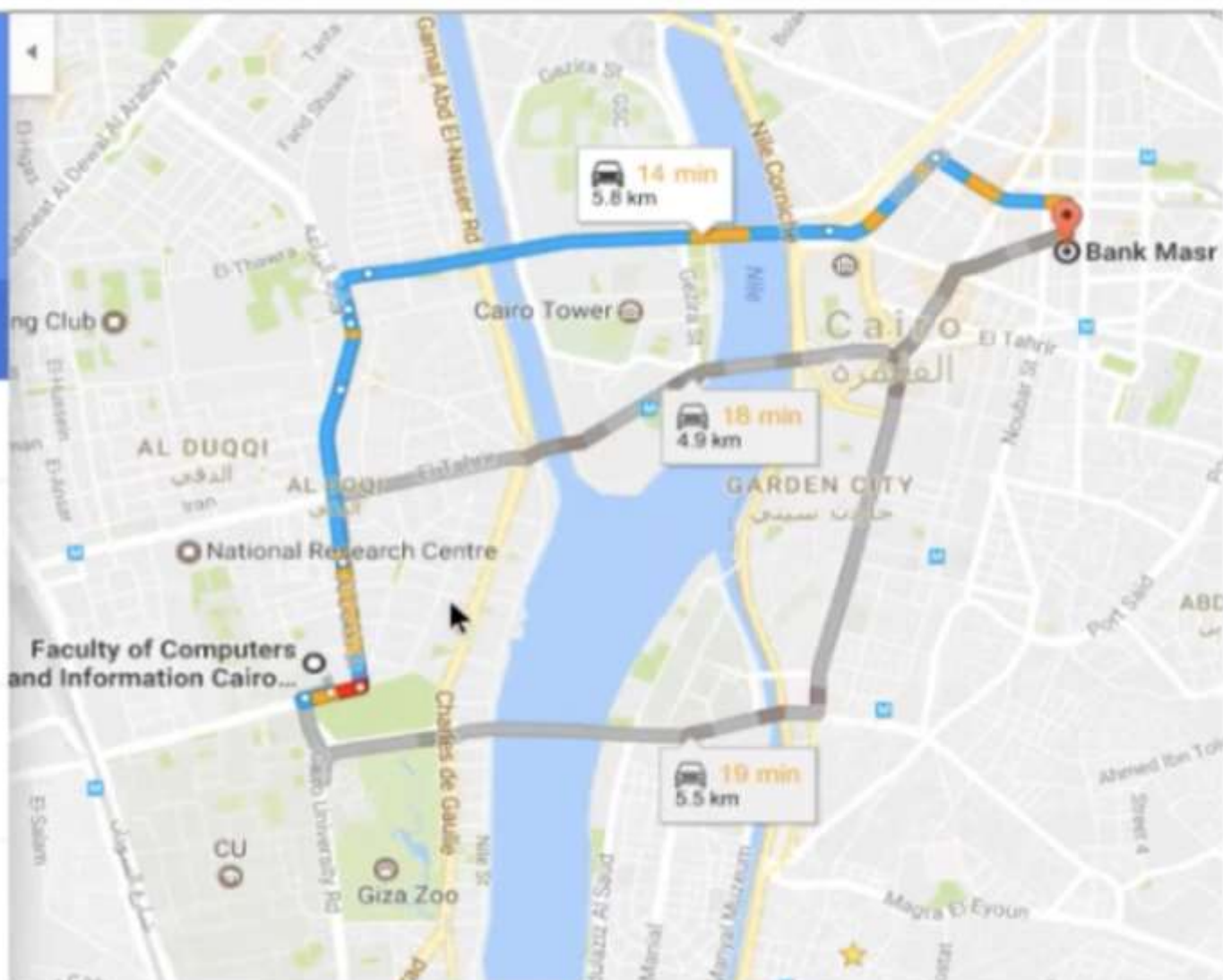
4.9 km

via Al Kasr Al Aini/Cairo - Al Wosta

The usual traffic

19 min

5.5 km



The common data structures

- From a project to another!
 - People noticed some data structures that **repeat** much
 - So, instead of each one keep **re**implementing them
 - They are **already** implemented for you to use
 - Hence, in real life projects, you build **more complex** data structures **based** on the common data structures
 - In a data structure **course**, we study these common data structures
 - Through this study, we learn how to **build** a data structure
 - Hence, we learn how to **use** them the commons properly
 - **Why learn them if already implemented?** Read above again slowly.

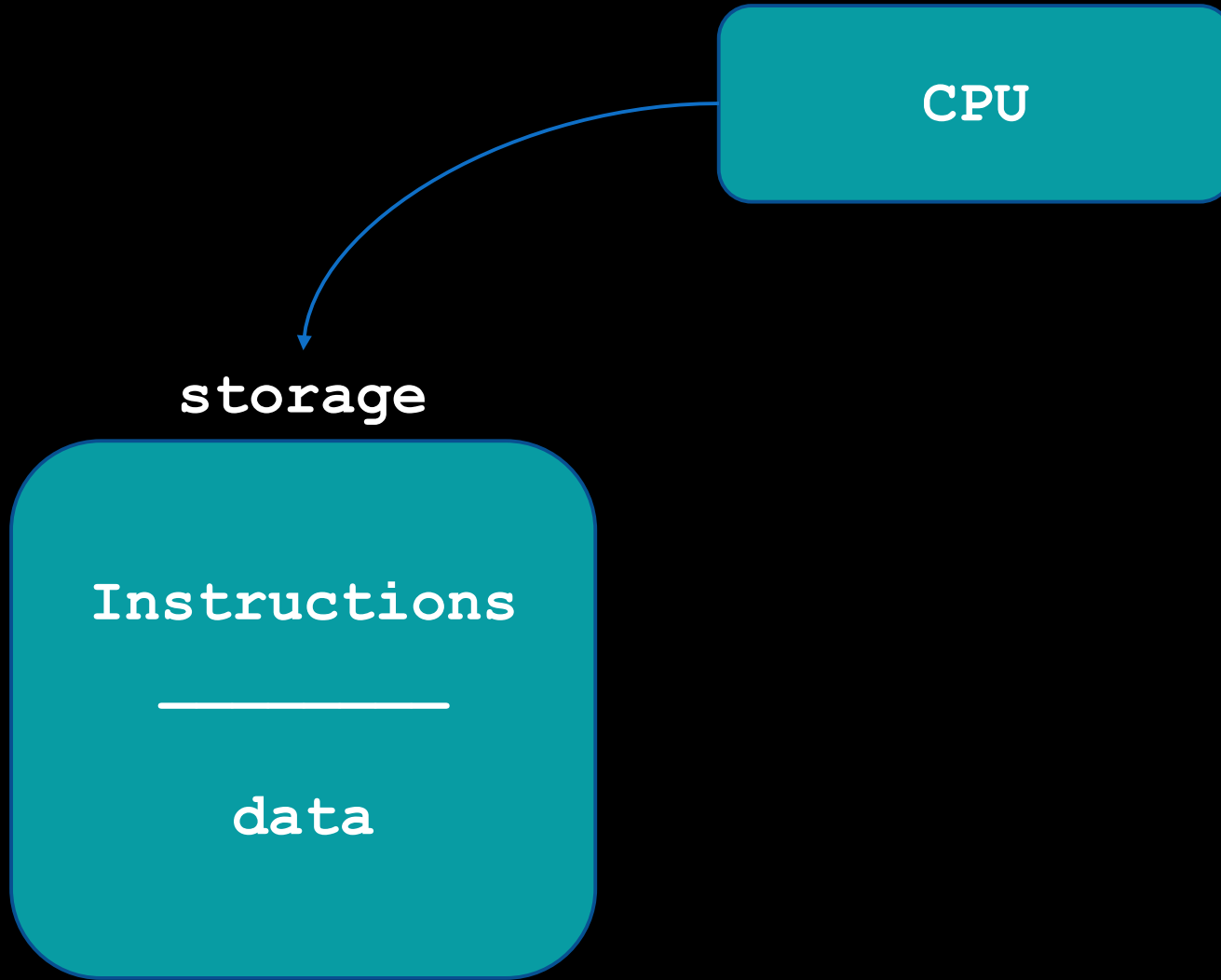
Program execution

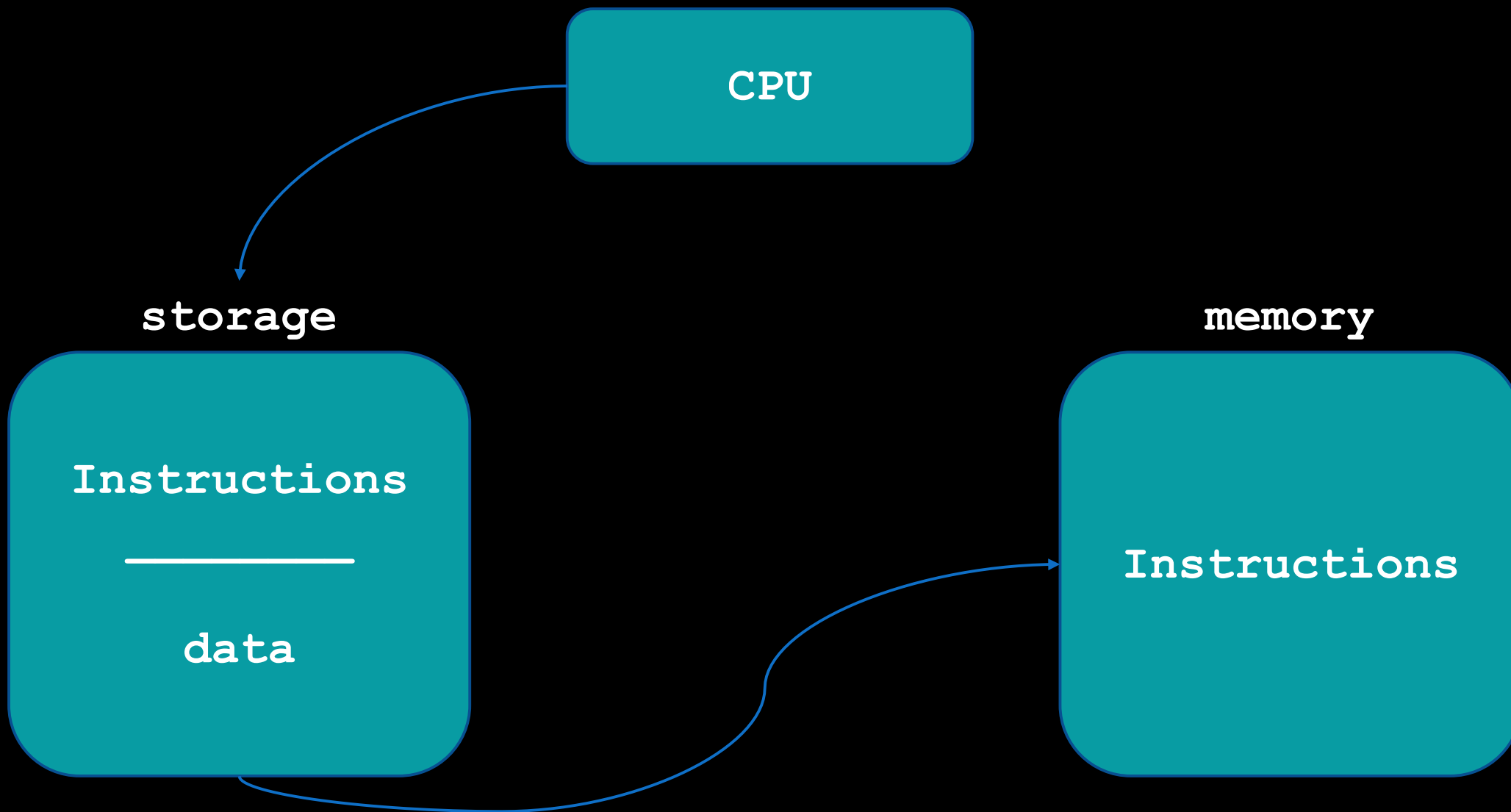
CPU

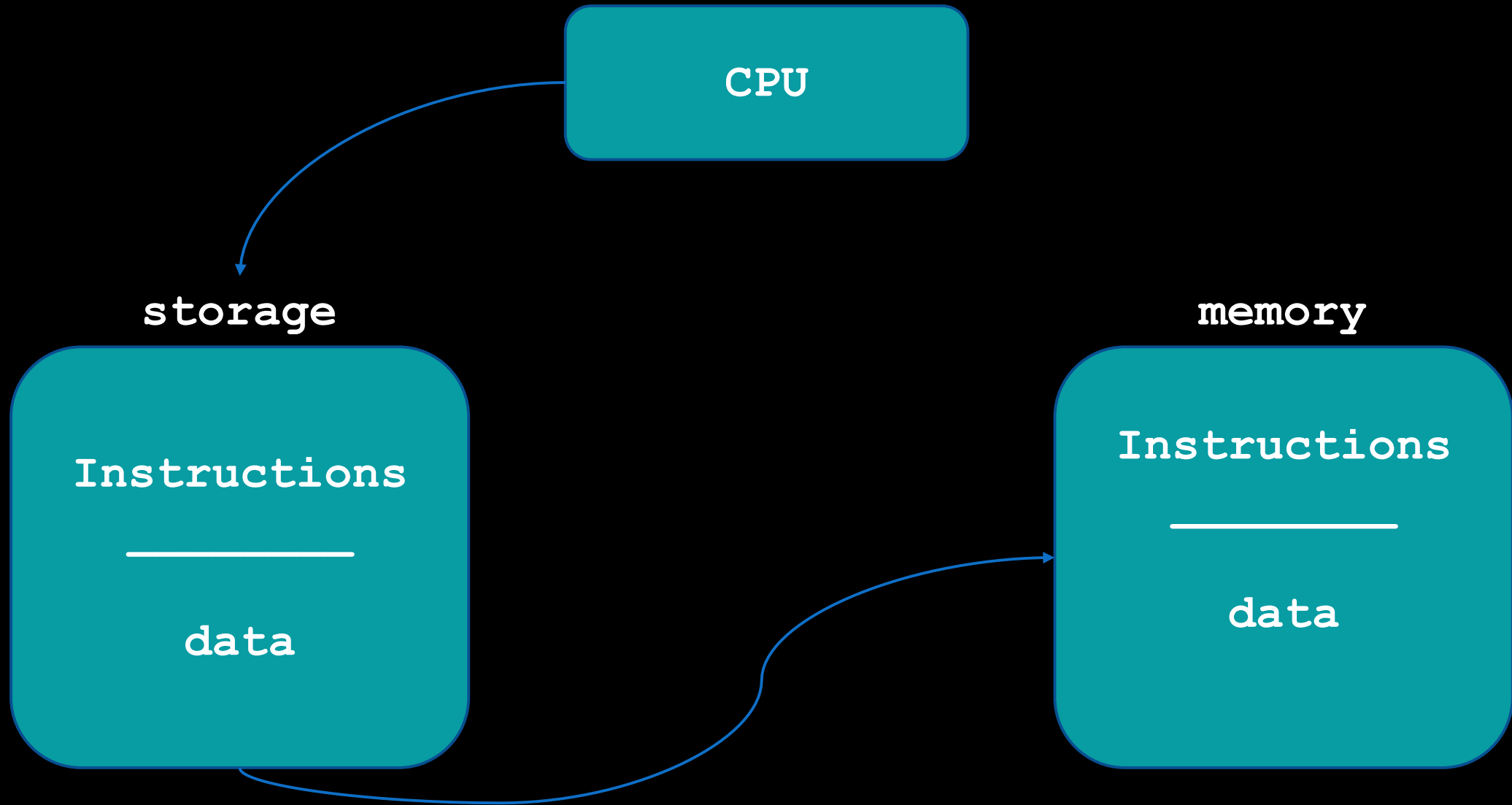
storage

Instructions

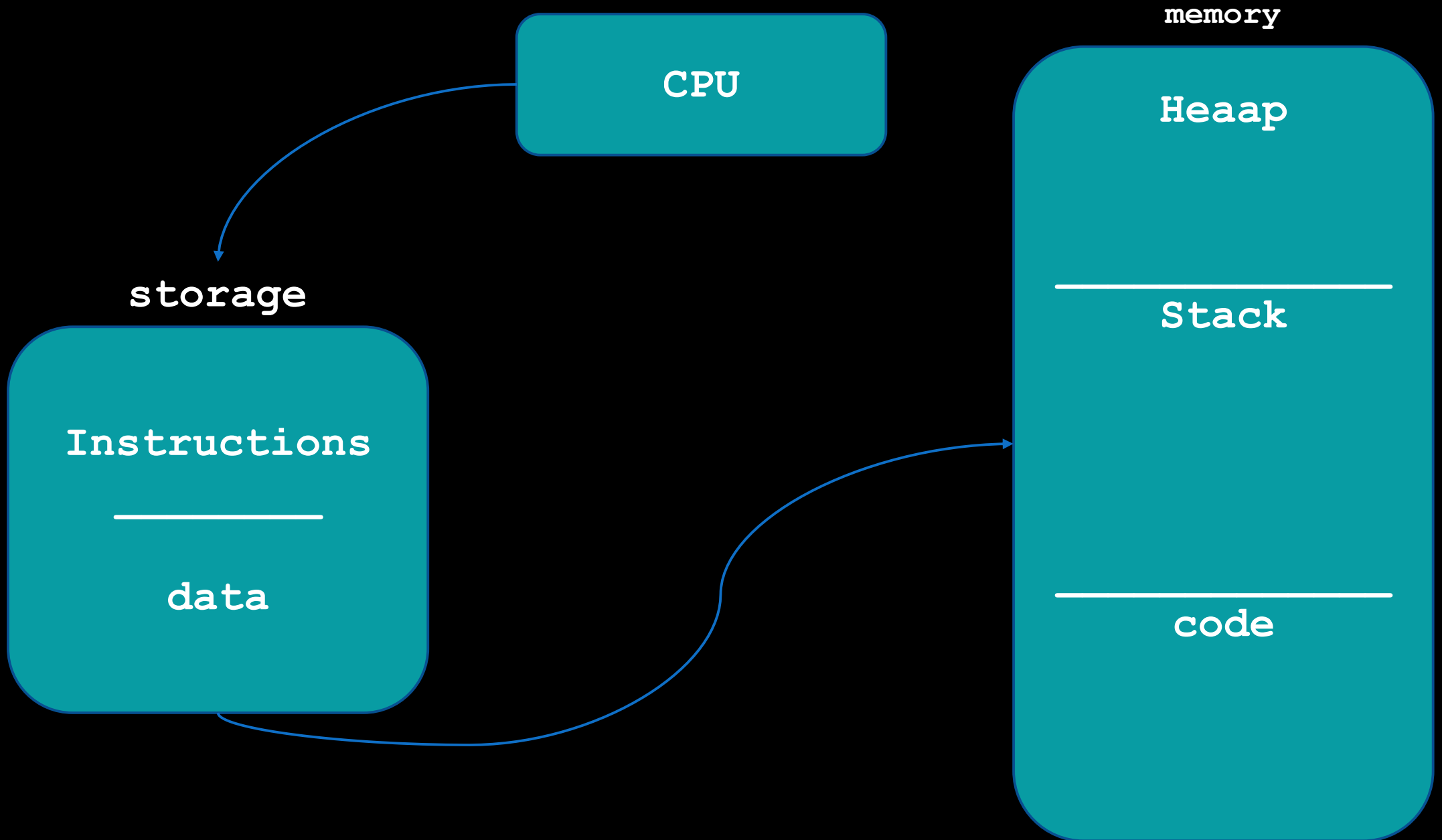
data







Memory Allocation



Static Allocation

CPU

storage

```
int main() {  
    int a, b;  
    Float avg;  
    avg = (a + b) / 2;  
    return 0;  
}
```

Heaap

Stack

a			avr	
	b			

code

CPU

storage

```
int main() {  
    int a, b;  
    Float avg;  
    avg = (a + b) / 2;  
    return 0;  
}
```

Heaap

Stack

8			8.5	
	9			

code

Static Allocation

- The computer knows exactly how much space an integer needs (usually 4 bytes) before the program even starts.
- The memory is allocated automatically when a function starts and is "popped" off (deleted) as soon as the function ends.

Dynamic Allocation

CPU

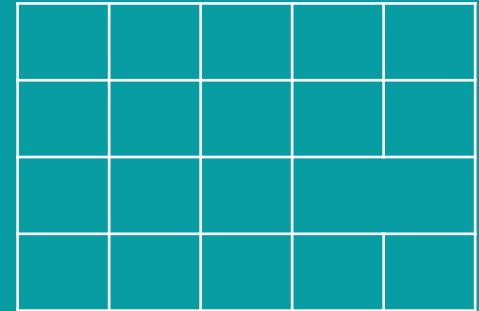
storage

```
int main() {  
    int i;  
    float *ptr;  
    cout << "Enter array  
size:";  
    cin >> i;  
    ptr = new float[i];  
    delete[] ptr;  
    return 0;  
}
```

Heaap



Stack



code

CPU

storage

```
int main() {  
    int i;  
    float *ptr;  
    cout << "Enter array  
size:";  
    cin >> i;  
    ptr = new float[i];  
    delete[] ptr;  
    return 0;  
}
```

Heaap

Stack

i				
			ptr	

code

Dynamic Allocation

- you don't know how much memory you need until the program is already running.
- The program "requests" a specific amount of space while it's running.
- This memory stays reserved until the program explicitly says "I'm done with this" (using delete or free). If you forget to do this, you get a memory leak.